

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Smart Lab Equipment & Slot Reservation Portal

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Course / Section	J
Date	28-08-2026

1. Lab Objective

The objective of Lab 2 is to use Jira to convert functional requirements from Lab 1 into Agile backlog items, create Epics and User Stories, prioritize and estimate the work, simulate Scrum sprints, and analyze progress using a Burndown Chart.

The Lab 2 handout lists the main learning outcomes as creating Epics and User Stories, prioritizing backlog items, assigning Fibonacci-based Story Points, applying Planning Poker as an estimation technique, simulating sprint progress, monitoring progress with a Burndown Chart, and reflecting on the sprint outcome.

2. Selected Requirement

Requirement selected for this lab:

FR-001 — Equipment Reservation

The system shall allow authenticated students to reserve calibrated equipment for a maximum of 2 hours, up to 7 days in advance, provided the equipment is available.

Requirement priority: High

3. Agile Breakdown

The selected functional requirement is represented by one Epic and four User Stories. The User Stories follow the lab's required format: As a [user role], I want [goal], so that [benefit].

User Story	Priority	Story Points	Status
Reserve Lab Equipment	High	5	To Do / Done as simulated
Check Equipment Availability	High	5	To Do / Done as simulated
Enforce 2-Hour Reservation Limit	High	3	To Do / Done as simulated
Enforce 7-Day Advance Booking	High	3	To Do / Done as simulated

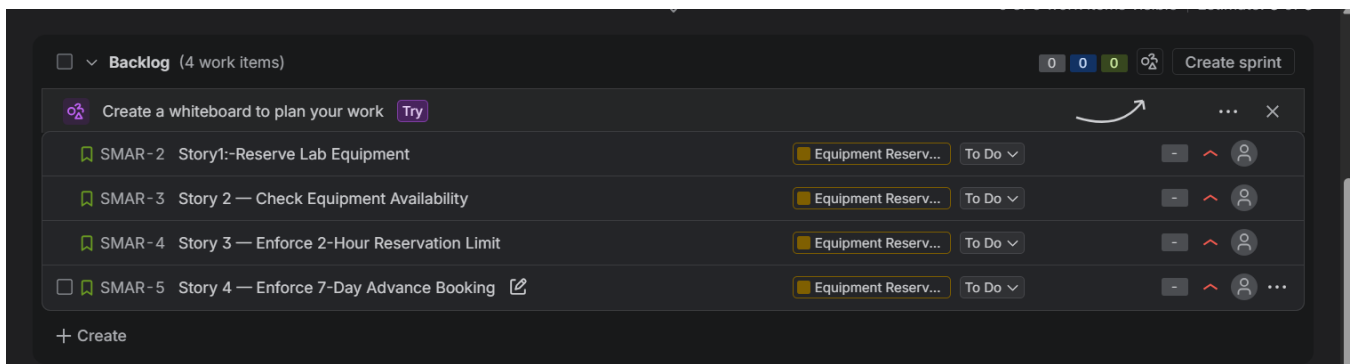
Total estimated effort: 16 Story Points.

4. Jira Backlog Creation

Steps followed:

1. Create a Scrum project in Jira using the Company-managed project type.
2. Open Backlog and create the Epic: Equipment Reservation.
3. Add the four User Stories under the Epic.
4. Use the required user-story format for each story.
5. Assign High priority to the stories based on the importance of the selected requirement.
6. Keep the stories in the backlog before starting the sprint.

Screenshot 1 — Backlog



5. Story Point Assignment

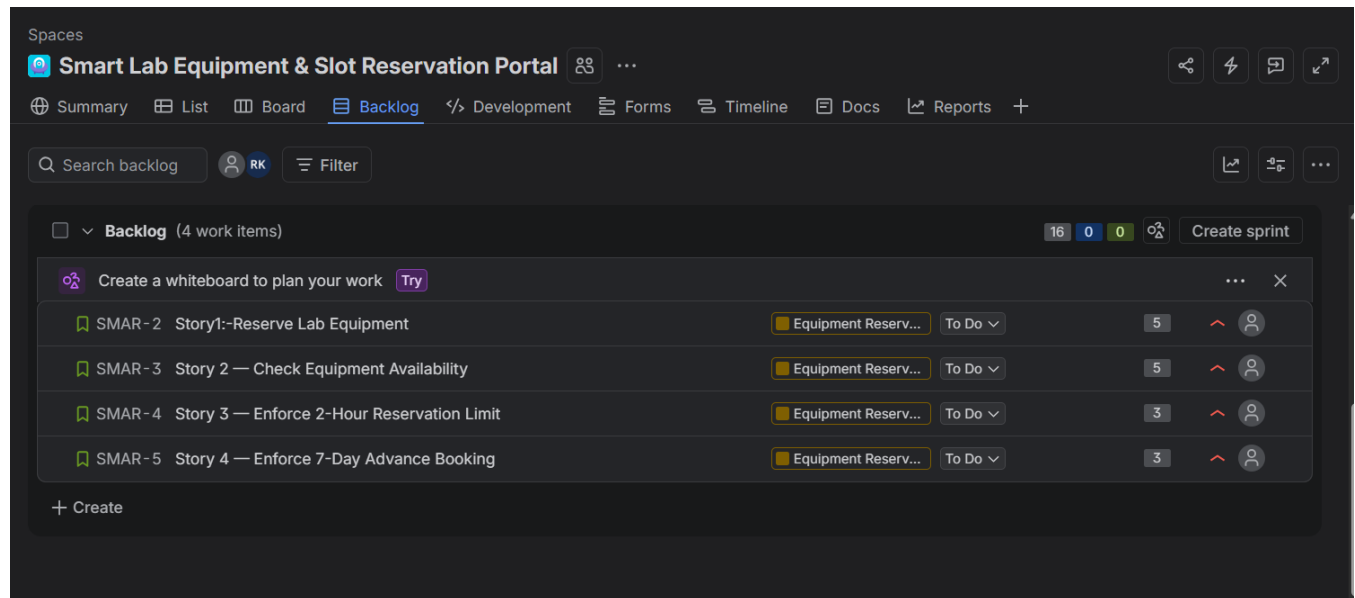
The Lab 2 handout instructs students to use the Fibonacci scale to estimate relative effort. Story Points represent overall effort, complexity, risk, and uncertainty rather than direct hours.

Planning Poker can be used by discussing a backlog item, independently estimating it, comparing estimates, and discussing differences until a reasonable estimate is reached.

Story	Estimate	Reason for Estimate
Reserve Lab Equipment	5	Main reservation functionality and associated validation.
Check Equipment Availability	5	Availability must be checked before confirmation.
Enforce 2-Hour Reservation Limit	3	Focused validation rule for maximum duration.
Enforce 7-Day Advance Booking	3	Focused validation rule for advance-booking window.

Total: 16 Story Points.

Screenshot 2 — Story Point Assignment



6. Sprint Planning and Simulation

The lab instructs students to create Sprint 1 with a one-week duration, place selected User Stories into the sprint, start the sprint, and simulate progress by moving work from To Do to In Progress to Done.

Sprint 1 planned content:

- Reserve Lab Equipment — 5 points
- Check Equipment Availability — 5 points
- Enforce 2-Hour Reservation Limit — 3 points
- Enforce 7-Day Advance Booking — 3 points

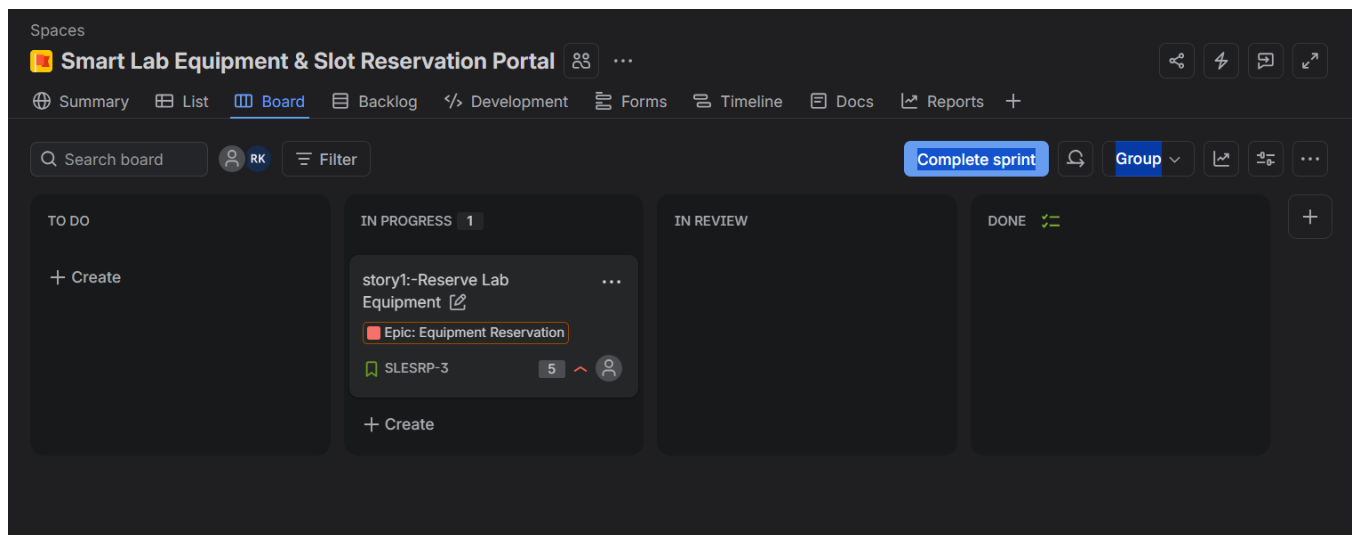
Sprint 1 total: 16 Story Points.

Suggested Sprint Goal: Implement the equipment reservation functionality with availability and booking-limit validation.

Sprint Simulation Procedure

7. Create Sprint 1 from the Backlog.
8. Move the selected stories into Sprint 1.
9. Set the sprint duration to one week.
10. Start the sprint.
11. Move stories from To Do to In Progress as work is simulated.
12. Move completed stories to Done.
13. Complete Sprint 1 when the planned work is finished.
14. Create and finish the second sprint as required by the lab handout.

Screenshot 3 — Active Sprint



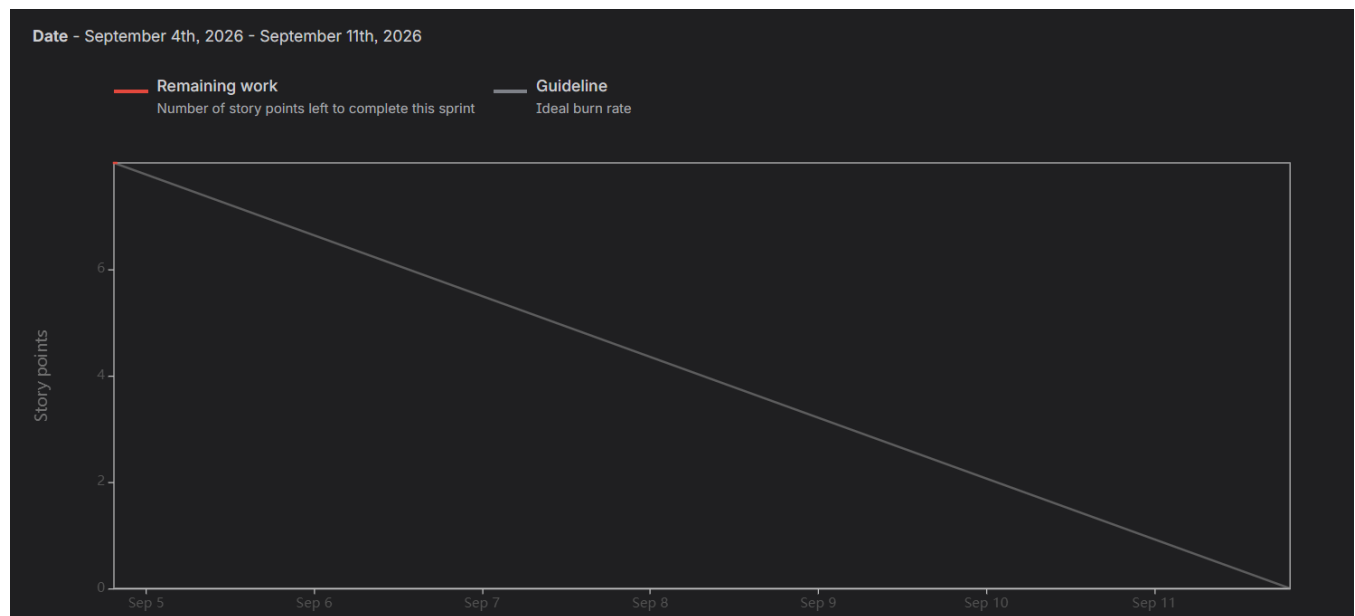
7. Burndown Chart

After the sprint simulation, open Jira Reports and select Burndown Chart. The chart shows work remaining over the sprint and helps evaluate whether the team is progressing toward completion.

Steps:

15. Open Reports in Jira.
16. Select Burndown Chart.
17. Select the relevant sprint if Jira asks for one.
18. Observe the change in remaining Story Points.
19. Use the chart to comment on early completion, delays, or inconsistencies.

Screenshot 4 — Burndown Chart



10. Final Review

The Lab 2 workflow is: Functional Requirement → Epic → User Stories → Prioritization → Story Points → Sprint 1 → Active Sprint Simulation → Sprint Completion → Sprint 2 → Burndown Chart → Reflection → Final PDF.